Sara Qamar

DEMO-2023-CS-0042 | Lumen Bay
sara.qamar@applicant.demo.invalid Portfolio
Search Intern | Coppercloud Metrics

Education

BSc Computer Science, Lumen Bay Institute of Computing, September 2023–July 2027
Expected graduation July 2027

Profile

Computer science student building web applications, data tools and search systems with Python, TypeScript and SQL.

Projects

FernFind Search Demo | Code

- Built a Python document search tool using SQLite FTS5 and BM25 ranking over fictional notes.
- Added pytest cases for empty queries, punctuation and deterministic result ordering.

Shelfmark Inventory Demo | Code

- Built a Flask inventory service with SQLite tables for products, stock movements and purchase records.
- Added pytest checks for stock adjustments and transaction rollback when an update fails.

QuayPulse Reports Demo | Code

- Built a Python pipeline that validates synthetic CSV records before loading them into DuckDB.
- Used SQL window functions to calculate rolling totals and identify duplicate event sequences, with pytest checks for invalid rows.

Glassline Delivery Lab Demo | Code

- Containerized a demonstration Flask service with Docker and documented its Linux startup procedure.
- Created a GitHub Actions workflow that runs tests and builds the container image.

Also built: Lantern Tasks

Technical Skills

Languages: Python
Engineering: SQLite, SQLite FTS5, BM25, pytest, SQL window functions, CSV validation, DuckDB, Docker
Backend: Flask
Data and SQL: SQL